Micah Winesberry

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EDUCATION

Yale University, New Haven, Connecticut — Expected Graduation: May 2030

- B.S. in Computer Science and Mathematics
- *Relevant Coursework:* Introductory Computer Science, Multivariable Calculus, Linear Algebra

Patrick F. Taylor Science & Technology Academy, Avondale, Louisiana — Graduation: May 2026

- *Awards:* National History Day Top 20 Individual Website Nationals (500,000+ Competitors), Patrick F. Taylor Scholarship ($100,000), Louisiana Science Fair 1st Place (Mathematics), Greater New Orleans Science Fair 1st Place (Systems Software), Mu Alpha Theta State 1st Place (Precalculus), State Rally 2nd Place (Calculus 2)

SKILLS

- *Languages & Tools:* Python, C, C++, Git, Linux, NumPy, Matplotlib
- *Technical:* Bluetooth Low Energy, Embedded Systems, Monte Carlo Simulation, Probabilistic Modeling, Algorithmic Optimization

PROJECTS & TECHNICAL EXPERIENCE

Flipper Zero Anki Remote | GitHub — Jun 2025 – Dec 2025

- Developed a C / Bluetooth Low Energy application for Flipper Zero that translates physical button inputs into configurable HID keyboard events.
- Implemented persistent preset management and SD-card storage for custom key mappings and device configuration.
- Reached **40,000+ global downloads** through the Flipper App Catalog.

Probabilistic Master Scheduler | GitHub — Aug 2025 – Jan 2026

- Developed a Python scheduler using Dirichlet-multinomial sampling, Dinic max-flow, MCMC simulated annealing, and Monte Carlo simulation to generate feasible high-school master schedules and predict student outcomes.
- Evaluated schedules using data from **600+ students** and 60+ course offerings, alongside survey data from 21 students.
- Compared simulated schedules with observed outcomes using paired t-tests and chi-square tests, identifying a gap between master scheduling and student placement procedures.

FSRS-5 Desired Retention Adjuster | GitHub — May 2024 – Jan 2025

- Implemented the FSRS-5 spaced-repetition algorithm in Python and introduced a parameter to simulate slower memory decay.
- Ran 495 experimental conditions to determine whether changes in decay rate could be approximated by adjusting desired retention.
- Used NumPy and Matplotlib to compare scheduling curves and visualize the resulting review intervals.

LEADERSHIP EXPERIENCE

Mu Alpha Theta (Math Honors Society) | Competition Prep Lead — Aug 2023 – May 2026

- Annotated 40+ practice exams and contributed to many individual, team, and interschool competition awards.
- Raised **$2,000** with exec team to enable the chapter's first state competition since 2013.

Patrick F. Taylor Chess Club | President — Aug 2022 – May 2026

- Reduced fees from **$150 to $5** through leadership restructuring, doubling membership from 15 to 30 students.
- Built a CustomTkinter application to manage in-club tournaments under our unique custom rules.

Science Olympiad | President and Founder — Sep 2025 – Feb 2026

- Founded and organized a full 15-member Science Olympiad team and managed a $1,500 preparation budget.
- Team placed **3rd overall** at the inaugural Tulane University Science Olympiad competition.